

Causal Offset Mean Calibration Pilot Campaign Results

Feedforward Transmission Error Baseline

Styled PDF edition generated from the canonical Markdown analysis report for improved readability, section hierarchy, and table presentation.

Overview

This report closes the `causal_offset_mean_calibration_pilot_2026_07_22` campaign. The campaign tested whether direct offset / curve-mean pressure gives a better next branch than continuing direct shape-threshold loss escalation.

The campaign used `polished_dataset` setpoint `Fw` data only and preserved both development roads requested for this stage:

- a time-windowed residual-offset GRU candidate;
- a non-windowed periodic MLP harmonic candidate.

The campaign completed remotely through manual recovery after the repository remote launcher and the user's terminal exited around transport handling. The remote runner produced `Completed 2` and `Failed 0`; the generated artifacts were synchronized back from the remote workstation.

Execution Summary

Item	Value
Campaign	causal_offset_mean_calibration_pilot_2026_07_22
Dataset	polished_dataset
Input mode	setpoints
Surface	Fw
Completed runs	2
Failed runs	0
Scalar winner	causal_offset_mean_periodic_mlp_harmonic_fw
Promotion	Not promoted
Required next gate	bounded TE Curve Verification Pipeline screen

Candidate Ranking

Rank	Family	Runtime Road	Validation MAE	Test MAE	Test RMSE	Params
1	causal_offset_mean_periodic_mlp_harmonic_fw	non-windowed harmonic MLP	0.001469	0.001277	0.001739	28,545
2	causal_offset_mean_gru_sequence_fw	time-windowed residual-offset GRU	0.002428	0.002100	0.002610	92,802

Metric Breakdown

Metric	Validation	Test
MLP pointwise loss	0.008758	0.006205
MLP centered curve shape loss	0.007616	0.005440
MLP curve offset loss	0.002398	0.002524
MLP curve amplitude loss	0.098801	0.070889
MLP sparse harmonic shape loss	0.000134	0.000094
GRU pointwise loss	0.021806	0.014675
GRU centered curve shape loss	0.019289	0.011694
GRU curve offset loss	0.002517	0.002981
GRU curve amplitude loss	0.106397	0.053888
GRU sparse harmonic shape loss	0.000443	0.000250
GRU base MAE	0.013305	0.013640
GRU residual offset mean abs	0.013476	0.014143

Pilot Comparison

Family	Surface	Validation MAE	Test MAE	Decision
periodic_gru_sequence_fw	Fw	0.001099	0.001101	Accepted model-development forward baseline remains stronger on scalar error.
shape_objective_periodic_mlp_harmonic_fw	Fw	0.001429	0.001236	Stronger scalar MLP reference, but failed bounded curve-first promotion.
causal_offset_mean_periodic_mlp_harmonic_fw	Fw	0.001469	0.001277	Campaign scalar winner; useful non-windowed result, not promoted.
periodic_mlp_harmonic_fw	Fw	0.001144	0.001326	Original non-windowed comparator; new MLP improves test MAE against this scalar baseline.
shape_first_distilled_periodic_mlp_harmonic_fw	Fw	0.001573	0.001420	New MLP improves scalar error against the prior distillation MLP.
shape_gate_loss_v2_checkpoint_selection_periodic_gru_sequence_fw	Fw	0.001983	0.001463	New MLP is stronger on scalar test MAE.
causal_offset_mean_gru_sequence_fw	Fw	0.002428	0.002100	Completed control, but not worth expanding in this exact profile.

Interpretation

The non-windowed harmonic MLP is the only useful result from this pilot. It improves the scalar test MAE against the original non-windowed `periodic_mlp_harmonic_fw` comparator and against the prior shape-first-distilled MLP, but it does not beat the strongest scalar shape-objective MLP reference and does not beat the accepted forward model-development GRU baseline.

The time-windowed residual-offset GRU did not work as a next road in this configuration. Its final test MAE is `0.002100 deg`, and the logged base MAE plus residual-offset magnitude show that the residual branch is making a large correction without becoming competitive. This branch should stay as completed evidence, not as the immediate expansion path.

The MLP result is not enough for promotion. The recent shape-objective pilot already showed that a strong scalar MLP can fail the bounded curve-first screen. This new winner therefore needs the same bounded `TE Curve Verification Pipeline` treatment before it can affect accepted model status.

Decision

Close the campaign as completed and do not promote either candidate.

The next repository step is a bounded `TE Curve Verification Pipeline` screen on `polished_dataset` setpoint `Fw` that compares:

- `causal_offset_mean_periodic_mlp_harmonic_fw` ;
- `causal_offset_mean_gru_sequence_fw` ;
- `polished_setpoints_periodic_gru_sequence_Fw` ;
- `polished_setpoints_periodic_mlp_harmonic_Fw` .

The screen may include `shape_objective_periodic_mlp_harmonic_Fw` as a scalar high-water reference, but promotion must still follow the multi-index curve-first policy rather than scalar `MAE` alone.

Operational Note

The campaign result is valid, but remote execution tooling needs follow-up before the next heavy campaign. The repository launcher and the user's terminal both exited during remote transport handling, and direct remote `conda run` returned a nonzero post-completion process code after writing successful campaign outputs. The campaign leaderboard is the canonical ranking surface for this closeout because the aggregate `output/registries/families/causal_offset_mean_calibration/leaderboard.yaml` artifact was not generated.

Primary Artifacts

- `output/training_campaigns/2026-07-22-18-09-27_causal_offset_mean_calibration_pilot_2026_07_22/campaign_leaderboard.yaml`
- `output/training_campaigns/2026-07-22-18-09-27_causal_offset_mean_calibration_pilot_2026_07_22/campaign_best_run.yaml`
- `output/training_campaigns/2026-07-22-18-09-27_causal_offset_mean_calibration_pilot_2026_07_22/campaign_best_run.md`
- `output/training_runs/causal_offset_mean_calibration/2026-07-22-18-15-50__te_causal_offset_mean_periodic_mlp_harmonic_fw__polished_setpoints/`
- `output/training_runs/causal_offset_mean_calibration/2026-07-22-18-09-27__te_causal_offset_mean_gru_sequence_fw__polished_setpoints/`
- `doc/reports/campaign_plans/cross_wave/causal_offset_mean_calibration/2026-07-22-17-42-11_causal_offset_mean_calibration_pilot_plan_report.md`
- `doc/reports/analysis/model_development_waves/wave_5_2/causal_offset_mean_calibration/[2026-07-22]/causal_offset_mean_calibration_pilot_model_report.md`